# Polynomial continued fractions with factorial reduction: seven hidden low-degree equivalences with proofs, a staged identification pipeline, and eight unidentified candidates

Vichien Fugsukjit
Independent researcher, Bangkok, Thailand
9chienvi9@gmail.com

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**Abstract**

We search polynomial continued fractions (PCFs) with quadratic partial denominators and quartic partial numerators for full factorial reduction (FR) and remove every case explained by known mechanisms (the family of Conjecture 1.3 of Ben David et al., deflation, rational or quadratic limits, Möbius transforms of standard constants and of H. Cohen's databases, and Euler-type PCFs detected through hypergeometric solutions of the recurrence). One PCF that passed all numerical tests is shown to be of Euler type in disguise (Theorem 1). Fifteen PCFs remained whose order-four operator is irreducible over $\mathbb{Q}(z)$ and whose limits match none of 21,984 constants at 1,000 digits. A further search for Möbius equivalences with PCFs of LOWER degree shows that seven of the fifteen are disguised degree-$(1,2)$ PCFs, i.e. ratios of contiguous Gauss hypergeometric functions (agreement to 170 digits). For ALL seven we give exact proofs by coboundaries (Theorem 2): the candidate is its partner, shifted by up to three steps, multiplied at each step by a constant times one linear factor of its own $b(n)$, and mixed by a polynomial gauge of degree at most six. This simple hiding mechanism is not detected by standard deflation tests. The remaining eight passed every test we could devise (Gamma products at 500 digits, 4,310 L-values of rational newforms, low-degree searches over about $10^9$ partner PCFs, monodromy, parameter deformation) and are published as candidates, not as new constants. We release the data, the code and a staged identification-and-proof pipeline that would have identified the seven on the first day.

## 1 Conventions

A PCF is $x = a(0) + \dfrac{b(1)}{a(1) + \dfrac{b(2)}{a(2) + \cdots}}$ with $a, b \in \mathbb{Z}[n]$; $P_n, Q_n$ satisfy $u_n = a(n)u_{n-1} + b(n)u_{n-2}$ with $P_{-1} = 1$, $P_0 = a(0)$, $Q_{-1} = 0$, $Q_0 = 1$. In matrix form $(u_{n-1}, u_n) = (u_{n-2}, u_{n-1})M(n)$ with $M(n) = \left(\begin{smallmatrix} 0 & b(n) \\ 1 & a(n) \end{smallmatrix}\right)$. Full FR means that $S_N = \frac{1}{N}\log(|Q_N|/\gcd(P_N, Q_N))$ stays bounded; numerically $|S_{800} - S_{200}| < 0.15$ (exact integers).

## 2 Search and filters

1. Brute force over about $2.3 \times 10^9$ PCFs ($\deg a = 2$, $b$ a product of four linear factors) with Möbius hashing against standard constants: 239 unidentified FR-candidates, 45 with full FR, 23 outside Conjecture 1.3 and a ${}_3F_2$ scan.

2. Extended search on the six $a(n)$ of these 23 with $b(n) = s\prod_{i=1}^{4}(k_i n + t_i)$: 5,274 PCFs with full FR.

3. Removed: Conjecture 1.3 incl. index shifts (846), slow convergence (15), rational or quadratic limits (313), deflation (3,684); Möbius clustering left 347 new clusters; a Cohen-database limit (1), $\pi\sqrt{3}$ (1), ${}_3F_2$ (2).

4. Euler type (order-one right factor of the recurrence over $\mathbb{Q}(n)$, ore_algebra): 61 + 1 cases (Section 3).

5. $L = \theta^2(\theta-1)^2 - z\,a(\theta+1)\theta^2 - z^2\,b(\theta+2)$ annihilates $\sum_n Q_n z^n/(n!)^2$; fifteen operators are irreducible over $\mathbb{Q}(z)$ (labels C0–C14, SHA-256 time-stamped before disclosure).

## 3 A hidden Euler-type example

**Theorem 1.** *For $a(n) = 4n^2 + 10n - 3$, $b(n) = -n^2(n+4)(3n-4)$: $P_n = -(n+1)!\prod_{m=0}^{n-1}(3m+2)\,(n^2+3n+3)$ and $x = 1/(S-1/3)$ with $S = -\frac{1}{24}\sum_{k\ge1}\frac{k!\,(k+2)(k+3)(k+4)}{(2/3)_k\,3^k\,(k^2+k+1)(k^2+3k+3)}$.*

*Proof sketch.* $r(n) = P_n/P_{n-1} = (n+1)(3n-1)(n^2+3n+3)/(n^2+n+1)$ satisfies $a(n)r(n-1)+b(n) = r(n)r(n-1)$ in $\mathbb{Q}(n)$; the Casoratian of $(P,Q)$ gives the series. Checked to 120 digits. □

With the same $a(n)$, $b(n) = (n+3)(2n-5)(2n-1)(3n-1)$ and $(n+1)(2n-3)(2n+1)(3n-5)$ give Möbius transforms of $\Gamma(1/3)^3\sqrt{3}/\pi$-type periods (250 digits); both operators factor as $[1,3]$.

## 4 Seven of the fifteen are disguised degree-$(1,2)$ PCFs

For each candidate $x$ we searched PCFs $y$ of lower degree (Batches 12–18: $\deg a \le 2$, $\deg b \le 2$, coefficient ranges up to $\pm60$ for $a$ and up to $\pm40$ or factored with $k \le 12, |t| \le 30$ for $b$; about $10^9$ partners) for a Möbius relation $y = (p+qx)/(r+sx)$, $|p|,\dots,|s| \le 8$, with planted, natural and random controls in every run. Every match was recomputed with exact recurrences to 160 digits.

| cand. | partner $a(n)$ | partner $b(n)$ | relation | $\lambda(n)$ | $s$ | $\deg V$ |
|---|---|---|---|---|---|---|
| C0 | $2n+3$ | $(3n+5)(5n+1)$ | $y = 3 - x$ | $2n+1$ | 1 | 3 |
| C10 | $n-3$ | $(2n+1)(3n+2)$ | $y = (8-6x)/(4-5x)$ | $3n+4$ | 1 | 3 |
| C11 | $n-3$ | $(3n-1)(4n+1)$ | $y = (8-3x)/(4+x)$ | $3n+1$ | 1 | 3 |
| C1 | $2n-1$ | $(3n-5)(5n-21)$ | $y = -(5+4x)/5$ | $2n+5$ | $-3$ | 3 |
| C5 | $6n-3$ | $9(3n-2)(5n-2)$ | $y = (x-3)/(-4-x)$ | $(2n+3)/3$ | $-3$ | 6 |
| C7 | $2n$ | $4(2n+3)(3n+1)$ | $y = -3x/(5-x)$ | $(3n+5)/2$ | $-2$ | 6 |
| C14 | $2n+8$ | $4(4n+1)(5n-7)$ | $y = (4+x)/6$ | $(3n+4)/2$ | $-2$ | 6 |

All seven relations are proven (Theorem 2); $\lambda$, $s$ and $V$ refer to the identity there; all $V$ matrices are in the released data. Degree-$(1,2)$ PCFs are ratios of contiguous ${}_2F_1$ functions; hence none of the seven is a new constant.

**Theorem 2.** *For each of the seven candidates there is a polynomial $2\times2$ matrix $V(n)$ with $\det V \not\equiv 0$ such that $M_1(n)\,V(n) = \lambda(n)\,V(n-1)\,M_2(n-s)$, with $\lambda$, $s$ as in the table; in every case $\lambda$ is a constant (1, 1/2 or 1/3) times a linear factor of the candidate's $b(n)$. For example, let $M_1$ be the matrix of candidate C0 ($a = 4n^2+10n-3$, $b = (2n-3)(2n+1)(3n+2)(5n-4)$) and $M_2$ that of its partner. Then $M_1(n)\,U(n) = (2n+1)\,U(n-1)\,M_2(n-1)$ with*

$$U(n) = \begin{pmatrix} \frac{10}{3}n^2 + \frac{7}{3}n - 2 & -10n^3 - \frac{41}{3}n^2 + 6n + \frac{5}{3} \\ -\frac{1}{3} & n \end{pmatrix},$$

$\det U(n) = -\frac{5}{9}(2n-1)(2n+1) \neq 0$ *for integers* $n$*; for C1,* $s=-3$*,* $\lambda = 2n+5$ *and*

$$V(n) = \begin{pmatrix} \frac{6}{5}n^2 + \frac{21}{5}n + \frac{18}{5} & 6n^3 + \frac{109}{5}n^2 + \frac{82}{5}n - \frac{21}{5} \\ \frac{1}{5} & n + \frac{8}{5} \end{pmatrix}, \qquad \det V(n) = \tfrac{11}{25}(2n+3)(2n+5).$$

*Proof and use.* Each identity is an equality of polynomial matrices and was verified by expanding it symbolically, by two independent scripts. Multiplying it for $n = 1, \ldots, N$ gives $A_1 M_1(1) \cdots M_1(N) = \Lambda_N \, G \, A_2 M_2(1) \cdots M_2(N-s) \, V(N)^{-1}$ with a scalar $\Lambda_N$ and a constant matrix $G$ built from $V(0)$, $A_i = \left(\begin{smallmatrix} 1 & a_i(0) \\ 0 & 1 \end{smallmatrix}\right)$ and at most three partner matrices; hence $x = (g_{11}y + g_{12})/(g_{21}y + g_{22})$. The exact $G$ obtained this way reproduces every relation of the table (numerical agreement better than $10^{-87}$). □

Mechanism: the candidate is its degree-$(1,2)$ partner shifted by $s$ steps and wrapped by $\lambda(n)$, mixed by a polynomial gauge; the partner's $b$ is (a scalar times) a product of two linear factors of the candidate's $b$, which turns the search into a cheap filter. The proofs were found by solving the identity as a linear system over $\mathbb{Q}$ for $V$, for $\lambda = \kappa\, p(n)/q(n)$ with $p, q$ built from linear factors of $b_1$, $b_2$ and their unit shifts; a determinant condition ($\det V(n)/\det V(n-1)$ must telescope) prunes the search exactly. Controls: C0 re-proven, a planted rational $\lambda$ recovered, no proof for C2 against a random partner. Earlier searches had missed C1, C5, C7, C14 because they used shifts $s \geq -2$ and integer $\kappa$ only.

**How common is the mechanism? (exploratory)** In a random sample of 1,065 full-FR PCFs of the extended search, the mechanism filter found a verified (60 digits) Möbius partner of degree $(1,2)$ for 168/291 of the Conjecture 1.3 family, 195/300 of the deflatable PCFs and 40/300 of the remaining ones ("survivors"). The pre-registered mismatch control (partners built from the factors of a DIFFERENT PCF) gave 12/150 verified matches, above the 2% limit we had set; the matches are genuine, which shows that the partners are generic rather than specific. These fractions are therefore lower bounds for Möbius equivalence to some degree-$(1,2)$ PCF, not a measure of the specific mechanism.

## 5 The eight unidentified candidates

| | $a(n)$ | $b(n)$ | $x$ | $\rho$ |
|---|---|---|---|---|
| C2 | $3n^2+8n+8$ | $2(2n+3)(3n+1)(3n+2)(3n+8)$ | $33.5147685956496535\ldots$ | $-187/42$ |
| C3 | $3n^2+8n+8$ | $2(2n+3)(3n-1)(3n+8)(5n+3)$ | $27.7112473245165813\ldots$ | $-187/45$ |
| C4 | $3n^2-2n+2$ | $2(2n-1)(3n-8)(3n-2)(5n-8)$ | $-9.6178657191277317\ldots$ | $3/5$ |
| C6 | $4n^2+9n+1$ | $(2n+3)(4n+1)(4n+9)(6n+7)$ | $39.4516159585089660\ldots$ | $-123/28$ |
| C8 | $6n^2+13n+5$ | $2(2n+1)(2n+5)(3n+4)(3n+7)$ | $36.8847177941179069\ldots$ | $-83/18$ |
| C9 | $6n^2+13n+5$ | $(3n-10)(3n-4)(3n+10)(5n+3)$ | $-41.3047171139817542\ldots$ | $-121/60$ |
| C12 | $4n^2+10n-3$ | $(2n-5)(2n+3)(3n+5)(5n-9)$ | $49.6956732085298041\ldots$ | $-39/20$ |
| C13 | $6n^2+13n+5$ | $2(2n+5)(2n+9)(4n+3)(5n-8)$ | $-54.0753445873853959\ldots$ | $-241/52$ |

$\rho$: non-integer local exponent of $L$ at its dominant singularity.

**Tests passed by all eight (none identified).** Not in Conjecture 1.3; not deflatable; limit not rational or quadratic; not Möbius ($|c| \leq 10^5$) of 36 standard constants, of the 635 limits of arXiv:2409.06086v4, of evaluable limits of arXiv:2607.06581, or of the 171 euler2ai limits; no hypergeometric solution; operator irreducible; PSLQ $[1, x, c, xc]$ against 21,984 constants at 1,000 digits; no match in the OEIS; connection coefficients not Γ-products at 150 and 480 digits (planted controls recovered, incl. a 53-element basis); no Möbius relation ($|c| \leq 10^8$) with 4,310 L-values of rational newforms of weight 2–4 and level $\leq 100$ (300 digits); no low-degree Möbius partner in the searches of

Section 4; traces of short words in the numerical monodromy are not algebraic of degree $\leq 4$ (controls: hypergeometric operators give 100% algebraic traces); shifting one linear factor of $b$ by its natural step gives no Möbius relation between neighbours of degree $\leq 4$.

## 6 Limitations

Novelty is not established: unpublished databases (e.g. the Ramanujan Library) were not available to us. For C12 the widest mechanism filter rejected 76 of 77 float matches; one partner ($a = 14n - 7$, $b = -54n^2 + 261n - 81$, complex characteristic roots) did not converge numerically and is unresolved. The negative results for the eight are limited to the searched boxes. Irreducibility is over $\mathbb{Q}(z)$ only. Full FR is a numerical criterion. The LMFDB has no search for decimal constants.

## Data and code

All plans were time-stamped on GitHub (github.com/vichienFF/ramanujan-explorer) before the corresponding runs; scripts, logs, results and the pipeline `identify_pipeline.py` (stages: basic constants, known structures, mechanism filter, low-degree search, constant libraries, operator analysis, coboundary proof; self-test included) are released with this preprint, together with the rational-$\lambda$ prover `b20_rational.py`, the seven proofs ($V$ matrices) and the independent check scripts.

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Computations used SageMath, ore_algebra, ramanujantools, mpmath, PARI/GP and Kaggle notebooks. The author used AI assistants (Anthropic Claude for code and analysis, OpenAI GPT as an independent checker); all claims are reproducible with the released code.

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